UNIVERSITY OF LEEDS

Weak Converse and Complement for Quantale-Enriched Profunctors

Ignacio Bellas Acosta[1] John G. Stell[1]

[1]School of Computer Science
University of Leeds

RAMiCS 2026

Overview

1. Motivation
2. Basic Notions
3. Stabilisation
4. Weak Converse
5. Weak Complement
6. Application to Mathematical Morphology

Motivation: Relation Algebras

- The full algebra of binary relations on a set X $Rel(X) = (\mathcal{P}X, \cup, \cap, \overline{(\cdot)}, \bot, \top, ;, id_X, (\cdot)^{\smile})$ are equipped with operations representing the converse and complement operations.

Motivation: Relation Algebras

- The full algebra of binary relations on a set X $Rel(X) = (\mathcal{P}X, \cup, \cap, \overline{(\cdot)}, \bot, \top, ;, id_X, (\cdot)^{\smile})$ are equipped with operations representing the converse and complement operations.
- Weakening relation [3] algebras have been proposed as a generalisation of relation algebras to account for monotone relations over posets.

Motivation: Relation Algebras

- The full algebra of binary relations on a set X $Rel(X) = (\mathcal{P}X, \cup, \cap, \overline{(\cdot)}, \bot, \top, ;, id_X, (\cdot)^{\smile})$ are equipped with operations representing the converse and complement operations.
- Weakening relation [3] algebras have been proposed as a generalisation of relation algebras to account for monotone relations over posets.
- Weakening relation algebras do not enjoy of a converse nor complement operations.

Motivation: Relation Algebras

- The full algebra of binary relations on a set X $Rel(X) = (\mathcal{P}X, \cup, \cap, \overline{(\cdot)}, \bot, \top, ;, id_X, (\cdot)^{\smile})$ are equipped with operations representing the converse and complement operations.
- Weakening relation [3] algebras have been proposed as a generalisation of relation algebras to account for monotone relations over posets.
- Weakening relation algebras do not enjoy of a converse nor complement operations.
- The usual converse and complement operation dualise the source and target poset.

Motivation: Relation Algebras

- The full algebra of binary relations on a set X $Rel(X) = (\mathcal{P}X, \cup, \cap, \overline{(\cdot)}, \bot, \top, ;, id_X, (\cdot)^{\smile})$ are equipped with operations representing the converse and complement operations.
- Weakening relation [3] algebras have been proposed as a generalisation of relation algebras to account for monotone relations over posets.
- Weakening relation algebras do not enjoy of a converse nor complement operations.
- The usual converse and complement operation dualise the source and target poset.
- SHAs [7] propose a solution by introducing two weak converse and complement operations.

Motivation: Mathematical Morphology

- Mathematical Morphology (M.M.): Body of techniques within image processing.

Motivation: Mathematical Morphology

- Mathematical Morphology (M.M.): Body of techniques within image processing.
- Originally using set theoretic tools to model binary images [4, 5] via two operations: Dilations and erosions.

Motivation: Mathematical Morphology

- Mathematical Morphology (M.M.): Body of techniques within image processing.
- Originally using set theoretic tools to model binary images [4, 5] via two operations: Dilations and erosions.
- A new wave has used M.M. tools to analyse transformations on graphical images [2].

Motivation: Mathematical Morphology

- Mathematical Morphology (M.M.): Body of techniques within image processing.
- Originally using set theoretic tools to model binary images [4, 5] via two operations: Dilations and erosions.
- A new wave has used M.M. tools to analyse transformations on graphical images [2].
- A relational approach using monotone relations on preorders has been proposed to account for binary graph M.M. [6, 8].

Motivation: Mathematical Morphology

- Mathematical Morphology (M.M.): Body of techniques within image processing.
- Originally using set theoretic tools to model binary images [4, 5] via two operations: Dilations and erosions.
- A new wave has used M.M. tools to analyse transformations on graphical images [2].
- A relational approach using monotone relations on preorders has been proposed to account for binary graph M.M. [6, 8].
- In [1] we proposed a relational framework to account for grey-scale graph M.M. using Ω-sup-fuzzy relations. However, it does not account for certain operations.

Quantaloids and Quantales

Definition

A quantaloid $\mathcal{R}$ is a **SupLat**-enriched category. A quantaloid morphism $f : \mathcal{R} \to \mathcal{R}'$ is a **SupLat**-enriched functor.

Quantaloids and Quantales

Definition

A quantaloid $\mathcal{R}$ is a **SupLat**-enriched category. A quantaloid morphism $f : \mathcal{R} \to \mathcal{R}'$ is a **SupLat**-enriched functor.

Definition

A quantaloid $\mathcal{R}$ is **Girard** if for all $r \in \mathcal{R}_0$ there is a $0_r : r \to r$ such that:

- $a \backslash 0_s = 0_r / a$,
- $0_s / (a \backslash 0_s) = a$.

Quantaloids and Quantales

Definition

A quantaloid $\mathcal{R}$ is a **SupLat**-enriched category. A quantaloid morphism $f : \mathcal{R} \to \mathcal{R}'$ is a **SupLat**-enriched functor.

Definition

A quantaloid $\mathcal{R}$ is **Girard** if for all $r \in \mathcal{R}_0$ there is a $0_r : r \to r$ such that:

- $a \backslash 0_s = 0_r / a$,
- $0_s / (a \backslash 0_s) = a$.

Definition

A quantale $\mathcal{Q}$ is a one object quantaloid.

Quantaloids and Quantales

Definition

A quantaloid $\mathcal{R}$ is a **SupLat**-enriched category. A quantaloid morphism $f : \mathcal{R} \to \mathcal{R}'$ is a **SupLat**-enriched functor.

Definition

A quantaloid $\mathcal{R}$ is **Girard** if for all $r \in \mathcal{R}_0$ there is a $0_r : r \to r$ such that:

- $a \backslash 0_s = 0_r / a$,
- $0_s / (a \backslash 0_s) = a$.

Definition

A quantale $\mathcal{Q}$ is a one object quantaloid.

We fix a **commutative** and **Girard** quantale $\mathcal{Q} = (\mathcal{Q}_*, \bullet, 1, 0)$.

Quantaloids and Quantales (Examples)

Example (Quantaloid)

The category **Rel** of sets and binary relations is a **Girard** quantaloid.

Example (Quantaloid)

The category **Ord** of preorders and monotone relations is a quantaloid.

Example (Quantale)

The two element Boolean Algebra $\mathbb{2}$ is a **commutative** and **Girard** quantale.

Example (Quantale)

Any MV-chain $Ł_n$ is a **commutative** and **Girard** quantale.

Quantale-Enriched Structures

Definition

A small $\mathcal{Q}$-category $\mathbb{C}$ is a set $\mathbb{C}_0$ and a hom-object $\mathbb{C}(-,-) : \mathbb{C}_0 \times \mathbb{C}_0 \to \mathcal{Q}_*$ where:

- $1 \leq \mathbb{C}(c,c)$,
- $\mathbb{C}(c',c'') \cdot \mathbb{C}(c,c') \leq \mathbb{C}(c,c'')$.

Let $\mathbb{C}^{op}$ be the $\mathcal{Q}$-category where $\mathbb{C}_0^{op} := \mathbb{C}_0$ and $\mathbb{C}^{op}(c,c') := \mathbb{C}(c',c)$.

Quantale-Enriched Structures

Definition

A small $\mathcal{Q}$-category $\mathbb{C}$ is a set $\mathbb{C}_0$ and a hom-object $\mathbb{C}(-,-) : \mathbb{C}_0 \times \mathbb{C}_0 \to \mathcal{Q}_*$ where:

- $1 \leq \mathbb{C}(c,c)$,
- $\mathbb{C}(c',c'') \cdot \mathbb{C}(c,c') \leq \mathbb{C}(c,c'')$.

Let $\mathbb{C}^{op}$ be the $\mathcal{Q}$-category where $\mathbb{C}_0^{op} := \mathbb{C}_0$ and $\mathbb{C}^{op}(c,c') := \mathbb{C}(c',c)$.

Definition

A $\mathcal{Q}$-functor $f : \mathbb{C} \to \mathbb{D}$ is a map $f : \mathbb{C}_0 \to \mathbb{D}_0$ where $\mathbb{C}(c,c') \leq \mathbb{D}(fc,fc')$.

Quantale-Enriched Structures

Definition

A small $\mathcal{Q}$-category $\mathbb{C}$ is a set $\mathbb{C}_0$ and a hom-object $\mathbb{C}(-,-) : \mathbb{C}_0 \times \mathbb{C}_0 \to \mathcal{Q}_*$ where:

- $1 \leq \mathbb{C}(c,c)$,
- $\mathbb{C}(c',c'') \cdot \mathbb{C}(c,c') \leq \mathbb{C}(c,c'')$.

Let $\mathbb{C}^{op}$ be the $\mathcal{Q}$-category where $\mathbb{C}_0^{op} := \mathbb{C}_0$ and $\mathbb{C}^{op}(c,c') := \mathbb{C}(c',c)$.

Definition

A $\mathcal{Q}$-functor $f : \mathbb{C} \to \mathbb{D}$ is a map $f : \mathbb{C}_0 \to \mathbb{D}_0$ where $\mathbb{C}(c,c') \leq \mathbb{D}(fc,fc')$.
Let $\mathbf{Cat}(\mathcal{Q})$ be the category of small $\mathcal{Q}$-categories and $\mathcal{Q}$-profunctors.

Definition

A $\mathcal{Q}$-profunctor $R : \mathbb{C} \nrightarrow \mathbb{D}$ is a function $R : \mathbb{D}_0 \times \mathbb{C}_0 \to \mathcal{Q}_*$ where:

- $\mathbb{C}(c,c') \cdot R(d,c) \leq R(d,c')$,
- $R(d,c) \cdot \mathbb{D}(d',d) \leq R(d',c)$.

Quantale-Enriched Structures (An Example)

- Suppose that $\mathcal{Q} = \mathbb{2}$

Quantale-Enriched Structures (An Example)

- Suppose that $\mathcal{Q} = \mathbb{2}$
- A small $\mathbb{2}$-category $\mathbb{C}$ is a set $\mathbb{C}_0$ and a hom-object $\mathbb{C}(-,-) : \mathbb{C}_0 \times \mathbb{C}_0 \to \mathbb{2}_*$ where:
  - $\top \leq \mathbb{C}(c, c)$,
  - $\mathbb{C}(c', c'') \wedge \mathbb{C}(c, c') \leq \mathbb{C}(c, c'')$.

  Hence a preorder!

Quantale-Enriched Structures (An Example)

- Suppose that $\mathcal{Q} = \mathbb{2}$
- A small $\mathbb{2}$-category $\mathbb{C}$ is a set $\mathbb{C}_0$ and a hom-object $\mathbb{C}(-,-) : \mathbb{C}_0 \times \mathbb{C}_0 \to \mathbb{2}_*$ where:
  - $\top \leq \mathbb{C}(c, c)$,
  - $\mathbb{C}(c', c'') \wedge \mathbb{C}(c, c') \leq \mathbb{C}(c, c'')$.

  Hence a preorder!
- A $\mathbb{2}$-functor $f : \mathbb{C} \to \mathbb{D}$ is a function $f : \mathbb{C}_0 \to \mathbb{D}_0$ such that if $\top \leq \mathbb{C}(c, c')$ then $\top \leq \mathbb{D}(fc, fc')$. Hence a monotone function!

Quantale-Enriched Structures (An Example)

- Suppose that $\mathcal{Q} = \mathbb{2}$
- A small $\mathbb{2}$-category $\mathbb{C}$ is a set $\mathbb{C}_0$ and a hom-object $\mathbb{C}(-,-) : \mathbb{C}_0 \times \mathbb{C}_0 \to \mathbb{2}_*$ where:
  - $\top \leq \mathbb{C}(c, c)$,
  - $\mathbb{C}(c', c'') \wedge \mathbb{C}(c, c') \leq \mathbb{C}(c, c'')$.

  Hence a preorder!
- A $\mathbb{2}$-functor $f : \mathbb{C} \to \mathbb{D}$ is a function $f : \mathbb{C}_0 \to \mathbb{D}_0$ such that if $\top \leq \mathbb{C}(c, c')$ then $\top \leq \mathbb{D}(fc, fc')$. Hence a monotone function!
- $\mathbf{Cat}(\mathbb{2}) \simeq \mathbf{PreOrd}$!

Quantale-Enriched Structures (An Example)

- Suppose that $\mathcal{Q} = \mathbb{2}$
- A small $\mathbb{2}$-category $\mathbb{C}$ is a set $\mathbb{C}_0$ and a hom-object $\mathbb{C}(-,-) : \mathbb{C}_0 \times \mathbb{C}_0 \to \mathbb{2}_*$ where:
  - $\top \leq \mathbb{C}(c, c)$,
  - $\mathbb{C}(c', c'') \wedge \mathbb{C}(c, c') \leq \mathbb{C}(c, c'')$.

  Hence a preorder!
- A $\mathbb{2}$-functor $f : \mathbb{C} \to \mathbb{D}$ is a function $f : \mathbb{C}_0 \to \mathbb{D}_0$ such that if $\top \leq \mathbb{C}(c, c')$ then $\top \leq \mathbb{D}(fc, fc')$. Hence a monotone function!
- $\mathbf{Cat}(\mathbb{2}) \simeq \mathbf{PreOrd}$!
- A $\mathbb{2}$-profunctor $R : \mathbb{C} \nrightarrow \mathbb{D}$ is a function $R : \mathbb{D}_0 \times \mathbb{C}_0 \to \mathbb{2}_*$ where:
  - $\mathbb{C}(c, c') \wedge R(d, c) \leq R(d, c')$,
  - $R(d, c) \wedge \mathbb{D}(d', d) \leq R(d', c)$.

  Therefore, it is a monotone relation between preorders!

Quantale-Enriched Profunctors

Definition

Let $\textsc{Prof}(\mathcal{Q})$ be the quantaloid where:

- $\textsc{Prof}(\mathcal{Q})_0 = \{\mathbb{C} \mid \mathbb{C}$ is a small $\mathcal{Q}$-category$\}$,
- $\textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) := \{R : \mathbb{C} \nrightarrow \mathbb{D} \mid R$ is a $\mathcal{Q}$-profunctor$\}$,
- $\otimes : \textsc{Prof}(\mathcal{Q})(\mathbb{D}, \mathbb{E}) \times \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) \to \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{E})$ where:

$$(S \otimes R)(e, c) := \bigvee_{d \in \mathbb{D}_0} R(d, c) \cdot S(e, d),$$

- $1_{\mathbb{C}} : \mathbb{C} \nrightarrow \mathbb{C}$ where $1_{\mathbb{C}}(c, c') = \mathbb{C}(c, c')$.

Quantale-Enriched Profunctors

Definition

Let $\textsc{Prof}(\mathcal{Q})$ be the quantaloid where:

- $\textsc{Prof}(\mathcal{Q})_0 = \{\mathbb{C} \mid \mathbb{C}$ is a small $\mathcal{Q}$-category$\}$,
- $\textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) := \{R : \mathbb{C} \nrightarrow \mathbb{D} \mid R$ is a $\mathcal{Q}$-profunctor$\}$,
- $\otimes : \textsc{Prof}(\mathcal{Q})(\mathbb{D}, \mathbb{E}) \times \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) \to \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{E})$ where:

$$(S \otimes R)(e, c) := \bigvee_{d \in \mathbb{D}_0} R(d, c) \cdot S(e, d),$$

- $1_{\mathbb{C}} : \mathbb{C} \nrightarrow \mathbb{C}$ where $1_{\mathbb{C}}(c, c') = \mathbb{C}(c, c')$.

Definition

Let $\langle \mathbb{C}, \mathbb{D} \rangle$ be the $\mathcal{Q}$-category where $\langle \mathbb{C}, \mathbb{D} \rangle_0 = \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D})$ and:

$$\langle \mathbb{C}, \mathbb{D} \rangle (R, R') := \bigwedge_{d,c} R(d, c) \backslash R'(d, c).$$

Operations on $\mathcal{Q}$-profunctors

Definition

Let $R : \mathbb{C} \nrightarrow \mathbb{D}$ be a $\mathcal{Q}$-profunctor. We define the:

Converse: $R^{\smile} : \mathbb{D}^{op} \nrightarrow \mathbb{C}^{op}$ where $R^{\smile}(c, d) := R(d, c)$,

Complement: $R^{\perp} : \mathbb{C}^{op} \nrightarrow \mathbb{D}^{op}$ where $R^{\perp}(d, c) := R(d, c) \backslash 0$,

Converse-Complement: $R^{\divideontimes} : \mathbb{D} \nrightarrow \mathbb{C}$ where $R^{\divideontimes}(c, d) := R(d, c) \backslash 0$.

Lemma

The maps:

Converse: $(\cdot)^{\smile} : \textsc{Prof}(\mathcal{Q}) \to \textsc{Prof}(\mathcal{Q})^{op}$ ($\mathbb{C} \mapsto \mathbb{C}^{op}$ and $R \mapsto R^{\smile}$),

Complement: $(\cdot)^{\perp} : \textsc{Prof}(\mathcal{Q}) \to \textsc{Prof}(\mathcal{Q})^{co}$ ($\mathbb{C} \mapsto \mathbb{C}^{op}$ and $R \mapsto R^{\perp}$),

Conv-Compl: $(\cdot)^{\divideontimes} : \textsc{Prof}(\mathcal{Q}) \to \textsc{Prof}(\mathcal{Q})^{coop}$ ($\mathbb{C} \mapsto \mathbb{C}$ and $R \mapsto R^{\divideontimes}$)

are involutive quantaloid morphisms.

Operations on $\mathcal{Q}$-profunctors

Definition

Let $R : \mathbb{C} \looparrowright \mathbb{D}$ be a $\mathcal{Q}$-profunctor. We define the:

Converse: $R^{\smile} : \mathbb{D}^{op} \looparrowright \mathbb{C}^{op}$ where $R^{\smile}(c, d) := R(d, c)$,

Complement: $R^{\perp} : \mathbb{C}^{op} \looparrowright \mathbb{D}^{op}$ where $R^{\perp}(d, c) := R(d, c) \backslash 0$,

Converse-Complement: $R^{\divideontimes} : \mathbb{D} \looparrowright \mathbb{C}$ where $R^{\divideontimes}(c, d) := R(d, c) \backslash 0$.

Lemma

Let $\mathbb{C}$ and $\mathbb{D}$ be small $\mathcal{Q}$-categories. Then, the functions:

Converse: $(\cdot)^{\smile} : \langle \mathbb{C}, \mathbb{D} \rangle \to \langle \mathbb{D}^{op}, \mathbb{C}^{op} \rangle$ $(R \mapsto R^{\smile})$,

Complement: $(\cdot)^{\perp} : \langle \mathbb{C}, \mathbb{D} \rangle \to \langle \mathbb{C}^{op}, \mathbb{D}^{op} \rangle^{op}$ $(R \mapsto R^{\perp})$,

Conv-Compl: $(\cdot)^{\divideontimes} : \langle \mathbb{C}, \mathbb{D} \rangle \to \langle \mathbb{D}, \mathbb{C} \rangle^{op}$ $(R \mapsto R^{\divideontimes})$

are $\mathcal{Q}$-isomorphisms.

Some Observations

Observation

- Converse/complement operations have the desired result that reverse/negate the original $\mathcal{Q}$-profunctor.
- Have *undesired* result that dualise the source and target categories.

Some Observations

Observation

- Converse/complement operations have the desired result that reverse/negate the original $\mathcal{Q}$-profunctor.
- Have *undesired* result that dualise the source and target categories.

Goal

Construct a *weaker* converse and complement operations that:

- Reverse/negate the original $\mathcal{Q}$-profunctor.
- **Preserve** the source and target categories.

Some Observations

Observation

- Converse/complement operations have the desired result that reverse/negate the original $\mathcal{Q}$-profunctor.
- Have *undesired* result that dualise the source and target categories.

Goal

Construct a *weaker* converse and complement operations that:

- Reverse/negate the original $\mathcal{Q}$-profunctor.
- **Preserve** the source and target categories.

Strategy

- Find a methodology that dualises the source and target categories of a profunctor (*Stabilisation*).
- Combine the *stabilisation* mechanism with the converse and complement operations.

Matrices

Definition

Let X and Y, a $\mathcal{Q}$-matrix $\Phi : X \not\to Y$ is a function $\Phi : Y \times X \to \mathcal{Q}$.

Matrices

Definition

Let X and Y, a $\mathcal{Q}$-matrix $\Phi : X \nrightarrow Y$ is a function $\Phi : Y \times X \to \mathcal{Q}$.

Definition

Let $\textsc{Matr}(\mathcal{Q})$ be the (large) quantaloid where:

- $\textsc{Matr}(\mathcal{Q})_0 := \{X \mid X \text{ is a set}\}$,
- $\textsc{Matr}(\mathcal{Q})(X, Y) := \{\Phi : X \nrightarrow Y \mid \Phi \text{ is a } \mathcal{Q}\text{-matrix}\}$,
- $\otimes : \textsc{Matr}(\mathcal{Q})(Y, X) \times \textsc{Matr}(\mathcal{Q})(X, Y) \to \textsc{Matr}(\mathcal{Q})(X, Z)$ where:

$$(\Psi \otimes \Phi)(z, x) := \bigvee_{y \in Y} \Phi(y, x) \cdot \Psi(z, y),$$

- $1_X : X \nrightarrow X$ where $1_X(x, x') := \begin{cases} 1 & x = x', \\ \bot & \text{otherwise.} \end{cases}$

Stabilisation

Proposition

The map $|\cdot| : \textsc{Prof}(\mathcal{Q}) \to \textsc{Matr}(\mathcal{Q})$ where:

- $|\mathbb{C}| = \mathbb{C}_0$ for all $\mathcal{Q}$-categories $\mathbb{C}$,
- $|\cdot|_{\mathbb{C},\mathbb{D}} : \textsc{Prof}(\mathcal{Q})(\mathbb{C},\mathbb{D}) \to \textsc{Matr}(\mathcal{Q})(\mathbb{C}_0,\mathbb{D}_0)$ $(R \mapsto R)$

is a lax quantaloid morphism.

Observation

Let $\mathbb{C}$ be a small $\mathcal{Q}$-category. As $|\mathbb{C}| = \mathbb{C}_0 = \mathbb{C}_0^{op} = |\mathbb{C}^{op}|$ the local functors $|\cdot|_{\mathbb{C},\mathbb{D}}$ and $|\cdot|_{\mathbb{C}^{op},\mathbb{D}^{op}}$ have the same target category.

$$\begin{tikzcd} \textsc{Prof}(\mathcal{Q})(\mathbb{C},\mathbb{D}) \arrow[r, shift left, "|\cdot|_{\mathbb{C},\mathbb{D}}"] & \textsc{Matr}(\mathcal{Q})(\mathbb{C}_0,\mathbb{D}_0) \arrow[l, shift left, "?"] \arrow[r, shift left, "?"] & \textsc{Prof}(\mathcal{Q})(\mathbb{C}^{op},\mathbb{D}^{op}) \arrow[l, shift left, "|\cdot|_{\mathbb{C}^{op},\mathbb{D}^{op}}"] \end{tikzcd}$$

Stabilisation

Lemma

Let $\mathbb{C}$ and $\mathbb{D}$ be two small $\mathcal{Q}$-categories. The local functor $|\cdot|_{\mathbb{C},\mathbb{D}} : \textsc{Prof}(\mathcal{Q})(\mathbb{C},\mathbb{D}) \to \textsc{Matr}(\mathcal{Q})(\mathbb{C}_0,\mathbb{D}_0)$ admits:

- *A left adjoint $\Diamond_{\mathbb{C},\mathbb{D}} : \textsc{Matr}(\mathcal{Q})(\mathbb{C}_0,\mathbb{D}_0) \to \textsc{Prof}(\mathcal{Q})(\mathbb{C},\mathbb{D})$ where $\Diamond_{\mathbb{C},\mathbb{D}}\Phi := 1_{\mathbb{D}} \otimes \Phi \otimes 1_{\mathbb{C}}$*
- *A right adjoint $\Box_{\mathbb{C},\mathbb{D}} : \textsc{Matr}(\mathcal{Q})(\mathbb{C}_0,\mathbb{D}_0) \to \textsc{Prof}(\mathcal{Q})(\mathbb{C},\mathbb{D})$ where $\Box_{\mathbb{C},\mathbb{D}}\Phi := 1_{\mathbb{D}} \blacktriangleright \Phi \blacktriangleleft 1_{\mathbb{C}}$*

Observation

$$\begin{tikzcd} \textsc{Prof}(\mathcal{Q})(\mathbb{C},\mathbb{D}) \arrow[r, shift left, "|\cdot|_{\mathbb{C},\mathbb{D}}"] \arrow[r, phantom, "\perp"] & \textsc{Matr}(\mathcal{Q})(\mathbb{C}_0,\mathbb{D}_0) \arrow[l, shift left, "\Box_{\mathbb{C},\mathbb{D}}"] \arrow[r, shift left, "\Diamond_{\mathbb{C}^{op},\mathbb{D}^{op}}"] \arrow[r, phantom, "\perp"] & \textsc{Prof}(\mathcal{Q})(\mathbb{C}^{op},\mathbb{D}^{op}) \arrow[l, shift left, "|\cdot|_{\mathbb{C}^{op},\mathbb{D}^{op}}"] \end{tikzcd}$$

Stabilisation

Corollary

Let $\mathbb{C}$ and $\mathbb{D}$ be small $\mathcal{Q}$-categories. Then, the maps:

- $\ell\text{-stab}^{\mathbb{C},\mathbb{D}} : \langle \mathbb{C}^{op}, \mathbb{D}^{op} \rangle \to \langle \mathbb{C}, \mathbb{D} \rangle$ where $\ell\text{-stab}^{\mathbb{C},\mathbb{D}}(R) = \Diamond_{\mathbb{C},\mathbb{D}} |R|$,
- $\text{r-stab}^{\mathbb{C},\mathbb{D}} : \langle \mathbb{C}^{op}, \mathbb{D}^{op} \rangle \to \langle \mathbb{C}, \mathbb{D} \rangle$ where $\text{r-stab}^{\mathbb{C},\mathbb{D}}(R) = \Box_{\mathbb{C},\mathbb{D}} |R|$,

are $\mathcal{Q}$-functors. Moreover $\ell\text{-stab}^{\mathbb{C},\mathbb{D}} \dashv \text{r-stab}^{\mathbb{C}^{op},\mathbb{D}^{op}}$ in $\mathbf{Cat}(\mathcal{Q})$.

Observation

$$\begin{tikzcd} \langle \mathbb{C}, \mathbb{D} \rangle \arrow[r, shift left, "\ell\text{-stab}"] \arrow[r, phantom, "\perp"] & \langle \mathbb{C}^{op}, \mathbb{D}^{op} \rangle \arrow[l, shift left, "\text{r-stab}"] \end{tikzcd}$$

Left and Right Converse

Definition

Let $R : \mathbb{C} \looparrowright \mathbb{D}$ be a $\mathcal{Q}$-functor. Let:

- The left converse $\circlearrowright R : \mathbb{D} \looparrowright \mathbb{C}$ where:

$$\circlearrowright R := \Diamond_{\mathbb{D},\mathbb{C}} |R^{\smile}| = \ell\text{-stab}^{\mathbb{D},\mathbb{C}}(R^{\smile}).$$

- The right converse $\circlearrowleft R : \mathbb{D} \looparrowright \mathbb{C}$ where:

$$\circlearrowleft R := (\Box_{\mathbb{C}^{op},\mathbb{D}^{op}} |R|)^{\smile} = \text{r-stab}^{\mathbb{C}^{op},\mathbb{D}^{op}}(R)^{\smile}.$$

Proposition

Let $\mathbb{C}, \mathbb{D}$ be two small $\mathcal{Q}$-categories. Then the monotone functions:

- $\circlearrowright : \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) \to \textsc{Prof}(\mathcal{Q})(\mathbb{D}, \mathbb{C})$ where $(R \mapsto \circlearrowright R)$,
- $\circlearrowleft : \textsc{Prof}(\mathcal{Q})(\mathbb{D}, \mathbb{C}) \to \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D})$ where $(R \mapsto \circlearrowleft R)$

and form an adjunction in **Pos**.

Left and Right Converse

Definition

Let $R : \mathbb{C} \looparrowright \mathbb{D}$ be a $\mathcal{Q}$-functor. Let:

- The left converse $\circlearrowright R : \mathbb{D} \looparrowright \mathbb{C}$ where:

$$\circlearrowright R := \Diamond_{\mathbb{D},\mathbb{C}} |R^{\smile}| = \ell\text{-stab}^{\mathbb{D},\mathbb{C}}(R^{\smile}).$$

- The right converse $\circlearrowleft R : \mathbb{D} \looparrowright \mathbb{C}$ where:

$$\circlearrowleft R := (\Box_{\mathbb{C}^{op},\mathbb{D}^{op}} |R|)^{\smile} = \text{r-stab}^{\mathbb{C}^{op},\mathbb{D}^{op}}(R)^{\smile}.$$

Proposition

$$\begin{tikzcd}
\textsc{Prof}(\mathcal{Q})(\mathbb{D}^{op}, \mathbb{C}^{op}) \arrow[r, shift left=2, "|\cdot|"] \arrow[r, phantom, "\perp"] \arrow[d, shift right, "(\cdot)^{\smile}"'] \arrow[d, phantom, "\simeq"] & \textsc{Matr}(\mathcal{Q})(\mathbb{D}_0, \mathbb{C}_0) \arrow[l, shift left=2, "\Box"] \arrow[d, shift right, "\Diamond"'] \arrow[d, phantom, "\vdash"] \\
\textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) \arrow[r, shift left=2, dashed, "\circlearrowright"] \arrow[r, phantom, "\perp"] \arrow[u, shift right, "(\cdot)^{\smile}"'] \arrow[u, phantom, "\simeq"] & \textsc{Prof}(\mathcal{Q})(\mathbb{D}, \mathbb{C}) \arrow[l, shift left=2, dashed, "\circlearrowleft"] \arrow[u, shift right, "|\cdot|"'] \arrow[u, phantom, "\vdash"]
\end{tikzcd}$$

Left and Right Converse

Proposition

Let $\mathbb{C}, \mathbb{D}$ be $\mathcal{Q}$-categories. The maps:

- $\circlearrowright : \langle \mathbb{C}, \mathbb{D} \rangle \to \langle \mathbb{D}, \mathbb{C} \rangle$ where $(R \mapsto \circlearrowright R)$,
- $\circlearrowleft : \langle \mathbb{D}, \mathbb{C} \rangle \to \langle \mathbb{C}, \mathbb{D} \rangle$ where $(R \mapsto \circlearrowleft R)$,

form an adjunction in $\mathbf{Cat}(\mathcal{Q})$.

Proposition

$$
\begin{tikzcd}
 & \langle \mathbb{D}^{op}, \mathbb{C}^{op} \rangle \arrow[dl, shift left, "(\cdot)^{\smile}"'] \arrow[dr, shift left, "\ell\text{-stab}"] \arrow[dl, phantom, "\simeq"] \arrow[dr, phantom, "\perp"] \\
\langle \mathbb{C}, \mathbb{D} \rangle \arrow[ur, shift left, "(\cdot)^{\smile}"] \arrow[rr, dashed, shift left, "\circlearrowright"] \arrow[rr, dashed, phantom, "\perp"] & & \langle \mathbb{D}, \mathbb{C} \rangle \arrow[ul, shift left, "\text{r-stab}"] \arrow[ll, dashed, shift left, "\circlearrowleft"]
\end{tikzcd}
$$

Left Complement

Definition

Let $R : \mathbb{C} \looparrowright \mathbb{D}$ be a $\mathcal{Q}$-profunctor. Let $\lrcorner R : \mathbb{C} \looparrowright \mathbb{D}$ be the left complement where $\lrcorner R := \Diamond_{\mathbb{C},\mathbb{D}} |R|^{\perp}$.

Proposition

Let $\mathbb{C}, \mathbb{D}$ be $\mathcal{Q}$-categories. The monotone function

$$\lrcorner : \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D})^{op} \to \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) \text{ where } (R \mapsto \lrcorner R)$$

is self adjoint arrow in **Pos**.

Proposition

Let $\mathbb{C}, \mathbb{D}$ be $\mathcal{Q}$-categories. The map $\lrcorner : \langle \mathbb{C}, \mathbb{D} \rangle^{op} \to \langle \mathbb{C}, \mathbb{D} \rangle$ where $(R \mapsto \lrcorner R)$ is self adjoint arrow in $\mathbf{Cat}(\mathcal{Q})$.

Left Complement

Definition

Let $R : \mathbb{C} \nrightarrow \mathbb{D}$ be a $\mathcal{Q}$-profunctor. Let $\lrcorner R : \mathbb{C} \nrightarrow \mathbb{D}$ be the left complement where $\lrcorner R := \Diamond_{\mathbb{C},\mathbb{D}} |R|^{\perp}$.

Proposition

$$\begin{tikzcd}
\textsc{Matr}(\mathcal{Q})(\mathbb{C}_0, \mathbb{D}_0)^{op} \arrow[r, shift left, "(\cdot)^{\perp}"] \arrow[r, phantom, "\simeq"] \arrow[d, shift left, "\Diamond"] \arrow[d, phantom, "\dashv"] & \textsc{Matr}(\mathcal{Q})(\mathbb{C}_0, \mathbb{D}_0) \arrow[l, shift left, "(\cdot)^{\perp}"] \arrow[d, shift left, "\Diamond"] \arrow[d, phantom, "\vdash"] \\
\textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D})^{op} \arrow[u, shift left, "|\cdot|"] \arrow[r, shift left, dashed, "\lrcorner"] \arrow[r, phantom, "\perp"] & \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) \arrow[l, shift left, dashed, "\lrcorner^{op}"] \arrow[u, shift left, "|\cdot|"]
\end{tikzcd}$$

Proposition

$$\begin{tikzcd}
\langle \mathbb{C}, \mathbb{D} \rangle^{op} \arrow[r, shift left, dashed, "\lrcorner"] \arrow[r, phantom, "\perp"] & \langle \mathbb{C}, \mathbb{D} \rangle \arrow[l, shift left, dashed, "\lrcorner^{op}"]
\end{tikzcd}$$

Right Complement

Definition

Let $R : \mathbb{C} \looparrowright \mathbb{D}$ be a $\mathcal{Q}$-profunctor. Let $\neg R : \mathbb{C} \looparrowright \mathbb{D}$ be the right complement where $\neg R := \Box_{\mathbb{C},\mathbb{D}}|R|^{\perp}$.

Proposition

Let $\mathbb{C}, \mathbb{D}$ be $\mathcal{Q}$-categories. The monotone function

$$\neg : \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) \to \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D})^{op} \text{ where } (R \mapsto \neg R)$$

is self adjoint arrow in **Pos**.

Proposition

Let $\mathbb{C}, \mathbb{D}$ be $\mathcal{Q}$-categories. The map $\neg : \langle \mathbb{C}, \mathbb{D} \rangle \to \langle \mathbb{C}, \mathbb{D} \rangle^{op}$ where $(R \mapsto \neg R)$ is self adjoint arrow in $\mathbf{Cat}(\mathcal{Q})$.

Right Complement

Definition

Let $R : \mathbb{C} \nrightarrow \mathbb{D}$ be a $\mathcal{Q}$-profunctor. Let $\neg R : \mathbb{C} \nrightarrow \mathbb{D}$ be the right complement where $\neg R := \square_{\mathbb{C},\mathbb{D}} |R|^{\perp}$.

Proposition

$$\begin{tikzcd}
\textsc{Matr}(\mathcal{Q})(\mathbb{C}_0, \mathbb{D}_0) \arrow[r, shift left, "(\cdot)^{\perp}"] \arrow[r, phantom, "\simeq"] \arrow[d, shift left, "\square"] \arrow[d, phantom, "\dashv"] & \textsc{Matr}(\mathcal{Q})(\mathbb{C}_0, \mathbb{D}_0)^{op} \arrow[l, shift left, "(\cdot)^{\perp}"] \arrow[d, shift left, "\square"] \arrow[d, phantom, "\vdash"] \\
\textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D}) \arrow[u, shift left, "|\cdot|"] \arrow[r, shift left, dashed, "\neg"] \arrow[r, phantom, "\perp"] & \textsc{Prof}(\mathcal{Q})(\mathbb{C}, \mathbb{D})^{op} \arrow[u, shift left, "|\cdot|"] \arrow[l, shift left, dashed, "\neg^{op}"]
\end{tikzcd}$$

Proposition

$$\begin{tikzcd}
\langle \mathbb{C}, \mathbb{D} \rangle \arrow[r, shift left, dashed, "\neg"] \arrow[r, phantom, "\perp"] & \langle \mathbb{C}, \mathbb{D} \rangle^{op} \arrow[l, shift left, dashed, "\neg^{op}"]
\end{tikzcd}$$

Combining all the Operations

Proposition

Let $\mathbb{C}, \mathbb{D}$ be small $\mathcal{Q}$-categories. The following diagrams commute:

$$\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\circlearrowright^{op}"] & \langle \mathbb{C}, \mathbb{D} \rangle^{op} \arrow[d, "(\cdot)^{\star}"] \\
\langle \mathbb{C}, \mathbb{D} \rangle \arrow[u, "(\cdot)^{\star}"] \arrow[r, "\circlearrowright"] & \langle \mathbb{D}, \mathbb{C} \rangle
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\circlearrowright^{op}"] & \langle \mathbb{C}, \mathbb{D} \rangle^{op} \arrow[d, "(\cdot)^{\star}"] \\
\langle \mathbb{C}, \mathbb{D} \rangle \arrow[u, "(\cdot)^{\star}"] \arrow[r, "\circlearrowright"] & \langle \mathbb{D}, \mathbb{C} \rangle
\end{tikzcd}$$

$$\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle \arrow[r, "\neg"] & \langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[d, "(\cdot)^{\star}"] \\
\langle \mathbb{C}, \mathbb{D} \rangle^{op} \arrow[u, "(\cdot)^{\star}"] \arrow[r, "\lrcorner"] & \langle \mathbb{C}, \mathbb{D} \rangle
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\lrcorner"] & \langle \mathbb{D}, \mathbb{C} \rangle \arrow[d, "(\cdot)^{\star}"] \\
\langle \mathbb{C}, \mathbb{D} \rangle \arrow[u, "(\cdot)^{\star}"] \arrow[r, "\neg"] & \langle \mathbb{C}, \mathbb{D} \rangle^{op}
\end{tikzcd}$$

Combining all the Operations

Proposition

Let $\mathbb{C}, \mathbb{D}$ be small $\mathcal{Q}$-categories. The following diagrams commute:

$$\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\smile^{op}"] \arrow[dr, "\lrcorner"'] & \langle \mathbb{C}, \mathbb{D} \rangle^{op} \arrow[d, "(\cdot)^{\star}"] \\
\langle \mathbb{C}, \mathbb{D} \rangle & \langle \mathbb{D}, \mathbb{C} \rangle
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\smile^{op}"] \arrow[dr, "\neg^{op}"'] & \langle \mathbb{C}, \mathbb{D} \rangle^{op} \arrow[d, "(\cdot)^{\star}"] \\
\langle \mathbb{C}, \mathbb{D} \rangle & \langle \mathbb{D}, \mathbb{C} \rangle
\end{tikzcd}$$

$$\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle \arrow[r, "\neg"] \arrow[dr, "\smile"'] & \langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[d, "(\cdot)^{\star}"] \\
\langle \mathbb{C}, \mathbb{D} \rangle^{op} & \langle \mathbb{C}, \mathbb{D} \rangle
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\lrcorner"] \arrow[dr, "\smile^{op}"'] & \langle \mathbb{D}, \mathbb{C} \rangle \arrow[d, "(\cdot)^{\star}"] \\
\langle \mathbb{C}, \mathbb{D} \rangle & \langle \mathbb{C}, \mathbb{D} \rangle^{op}
\end{tikzcd}$$

Combining all the Operations

Proposition

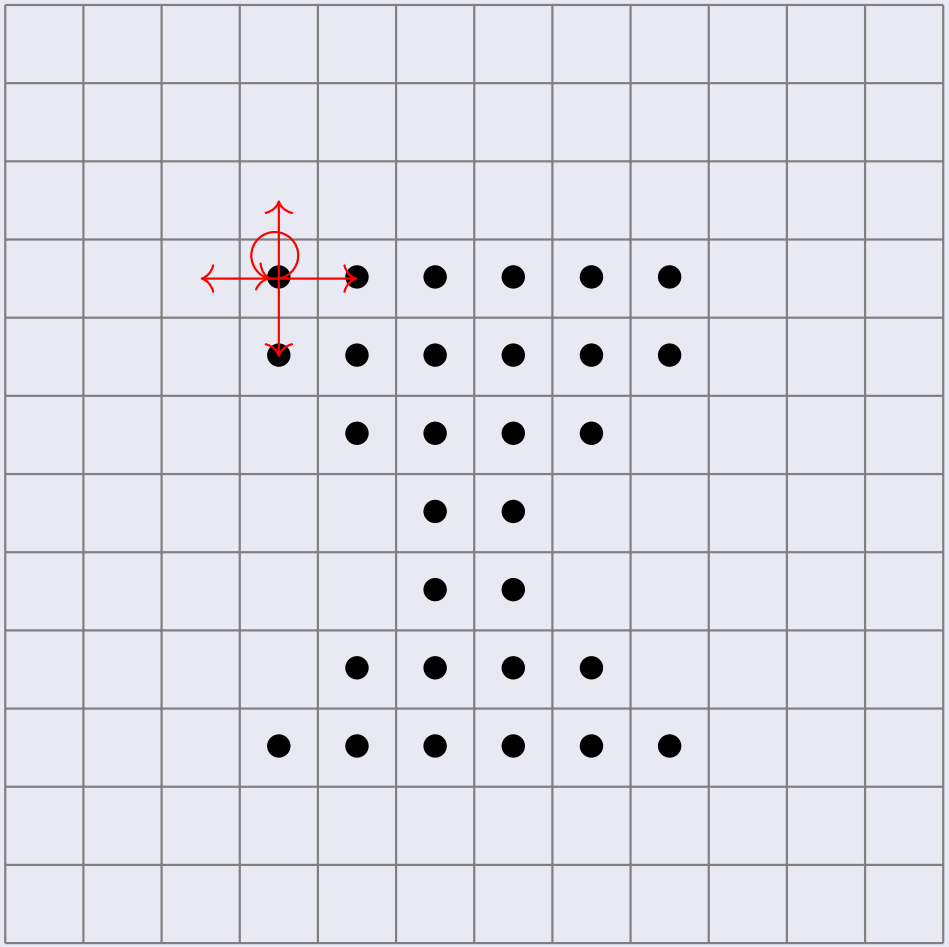
Let $\mathbb{C}, \mathbb{D}$ be small $\mathcal{Q}$-categories. The following diagrams commute:

$$\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\smile^{op}"] & \langle \mathbb{C}, \mathbb{D} \rangle^{op} \\
\langle \mathbb{C}, \mathbb{D} \rangle \arrow[u, "(\cdot)^{\star}"] \arrow[ur, "\neg"'] & \langle \mathbb{D}, \mathbb{C} \rangle
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\frown^{op}"] & \langle \mathbb{C}, \mathbb{D} \rangle^{op} \\
\langle \mathbb{C}, \mathbb{D} \rangle \arrow[u, "(\cdot)^{\star}"] \arrow[ur, "\lrcorner^{op}"'] & \langle \mathbb{D}, \mathbb{C} \rangle
\end{tikzcd}$$

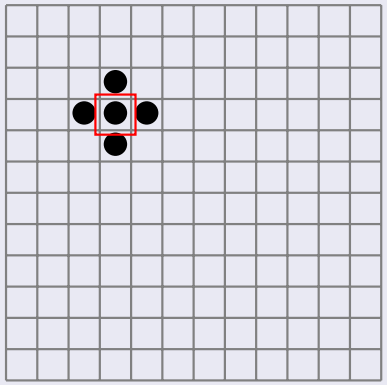
$$\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle \arrow[r, "\neg"] & \langle \mathbb{D}, \mathbb{C} \rangle^{op} \\
\langle \mathbb{C}, \mathbb{D} \rangle^{op} \arrow[u, "(\cdot)^{\star}"] \arrow[ur, "\smile^{op}"'] & \langle \mathbb{C}, \mathbb{D} \rangle
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \mathbb{D}, \mathbb{C} \rangle^{op} \arrow[r, "\lrcorner"] & \langle \mathbb{D}, \mathbb{C} \rangle \\
\langle \mathbb{C}, \mathbb{D} \rangle \arrow[u, "(\cdot)^{\star}"] \arrow[ur, "\frown"'] & \langle \mathbb{C}, \mathbb{D} \rangle^{op}
\end{tikzcd}$$

Mathematical Morphology

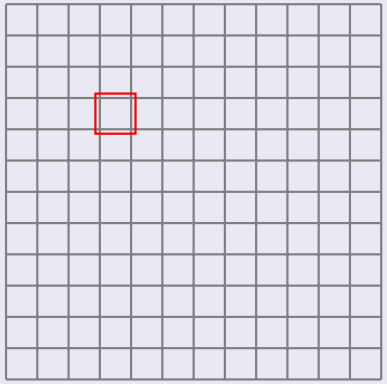
Image



Dilation

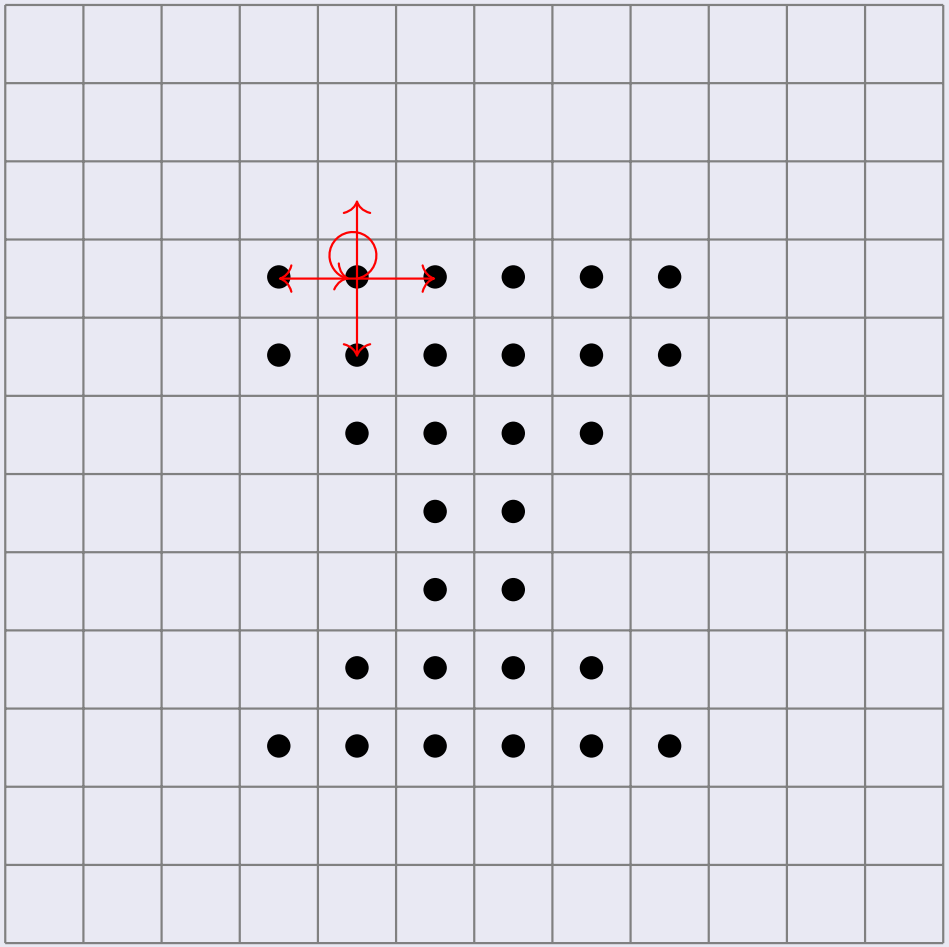


Erosion

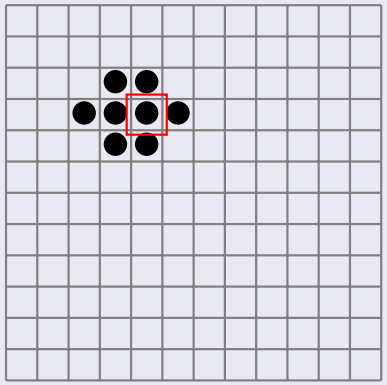


Mathematical Morphology

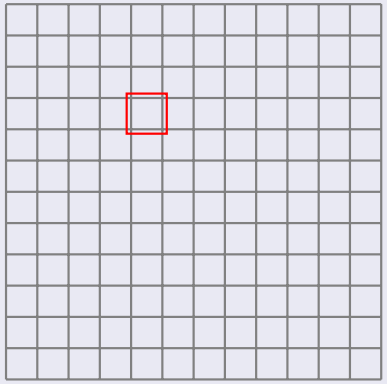
Image



Dilation

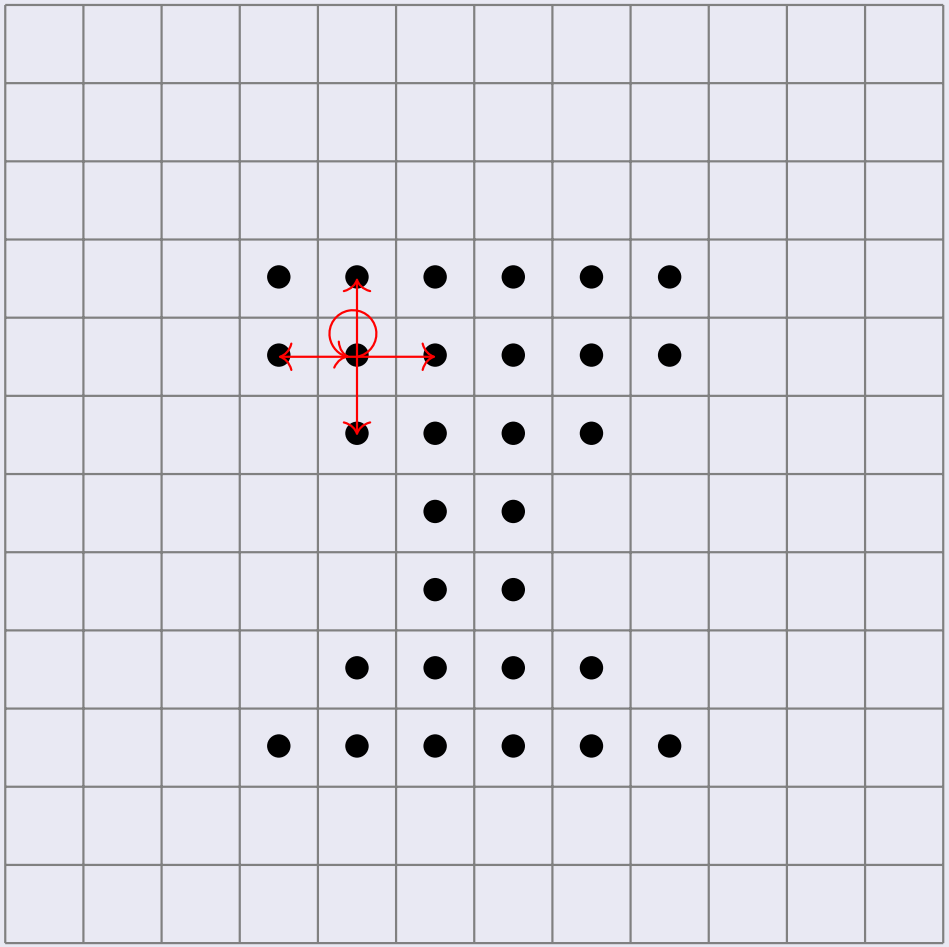


Erosion

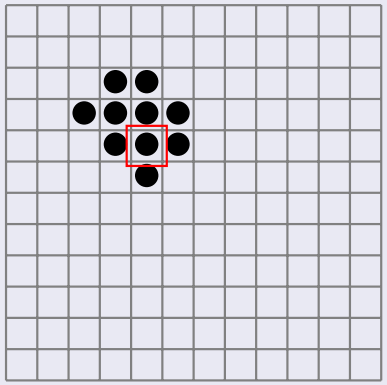


Mathematical Morphology

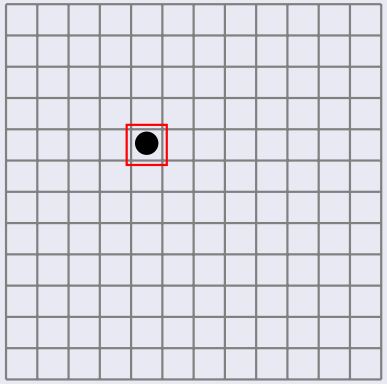
Image



Dilation

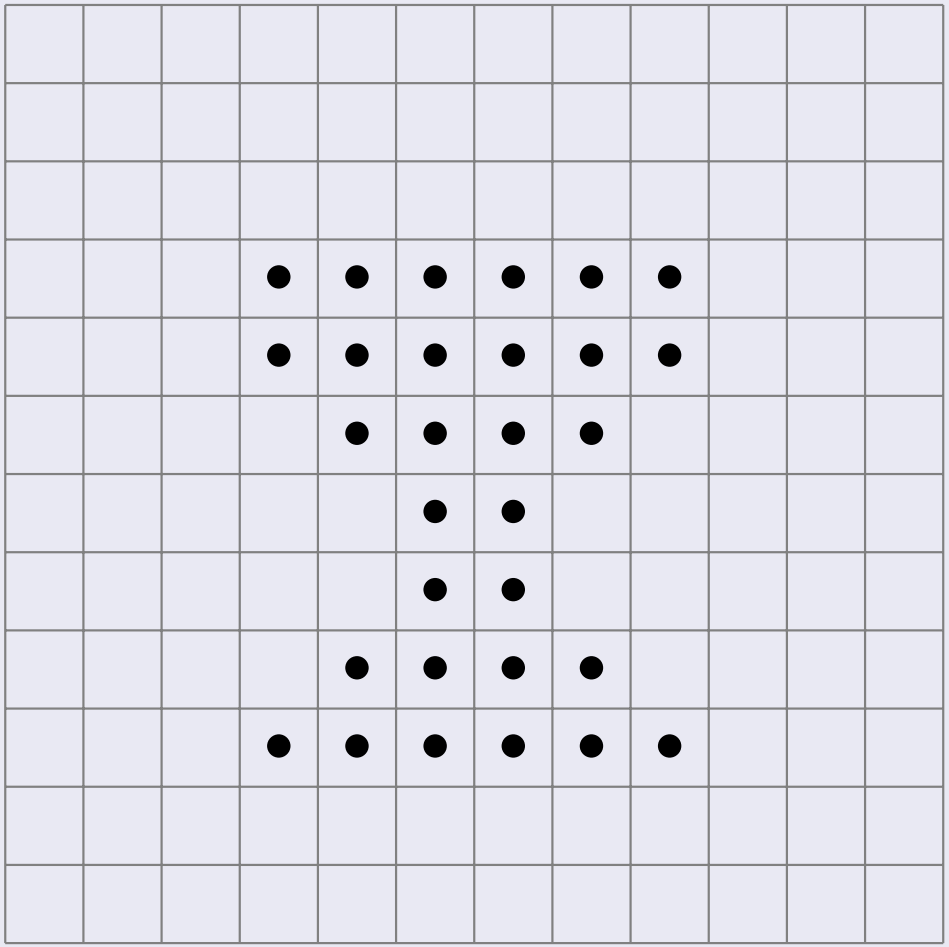


Erosion

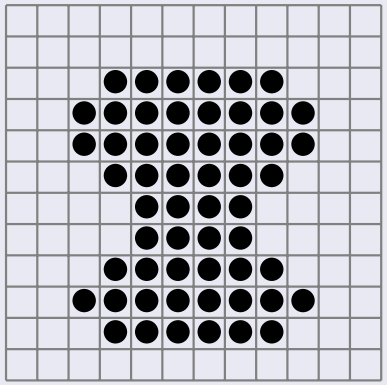


Mathematical Morphology

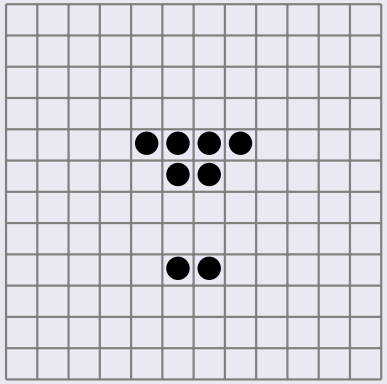
Image



Dilation



Erosion



Mathematical Morphology

Definition

Let X be a set. A dilation is a sup-lattice morphism $\delta : \mathcal{P}X \to \mathcal{P}X$. Dually, an erosion is an inf-lattice morphism $\varepsilon : \mathcal{P}X \to \mathcal{P}X$.

Let $\mathcal{Q}_\delta(X)$ be the quantale of dilations $\mathcal{P}X \to \mathcal{P}X$.

Definition

Let X be a set and $R \subseteq X \times X$. Let $- \oplus R : \mathcal{P}X \to \mathcal{P}X$ where:

$$A \oplus R := \{x \in X \mid \exists a \in A : aRx\}.$$

Moreover, let $- \ominus R : \mathcal{P}X \to \mathcal{P}X$ be the function where:

$$A \ominus R := \{x \in X \mid \forall a \in X : xRa \text{ implies } a \in A\}.$$

Let $\mathbf{Rel}(X)$ be the quantale of binary relations on X.

Mathematical Morphology

Lemma

Let X be a set. Then $\mathcal{Q}_\delta(X) \simeq \mathbf{Rel}(X) \simeq \mathcal{Q}_\varepsilon(X)^{op}$.

Lemma

Let X be a set and $R \subseteq X \times X$. Then the following diagrams commute:

$$\begin{tikzcd} \mathcal{P}X^{op} \arrow[r, "-\ominus R"] \arrow[d, "\overline{(\cdot)}"'] & \mathcal{P}X \\ \mathcal{P}X \arrow[r, "-\oplus R^{\smile}"] & \mathcal{P}X \arrow[u, "\overline{(\cdot)}"'] \end{tikzcd} \qquad \begin{tikzcd} \mathcal{P}X \arrow[r, "-\oplus R"] \arrow[d, "\overline{(\cdot)}"'] & \mathcal{P}X \\ \mathcal{P}X^{op} \arrow[r, "-\ominus R^{\smile}"] & \mathcal{P}X^{op} \arrow[u, "\overline{(\cdot)}"'] \end{tikzcd}$$

Goal

Generalise the M.M. framework to account for grey-scale graphical images.

Mathematical Morphology

Solution

Use the theory developed on quantale-enriched structures:

- Quantales: Formalisation of colour-scales:
  - Boolean Algebra $\mathbb{2}$: Model black-white colour scale.
  - MV-chains $Ł_n$: Model grey-scale.
- Enrichment: Formalise graphical structure.

Definition

Let $\mathcal{H} = (V, E)$ be a hypergraph. Let $\mathbb{H}$ be the $\mathcal{Q}$-category where $\mathbb{H}_0 = V \uplus E$ and:

$$\mathbb{H}(a, b) := \begin{cases} 1 & a = b, \\ 1 & a \in E, b \in V \text{ and } b \in a, \\ \bot & \text{otherwise.} \end{cases}$$

Mathematical Morphology

Question 1

How do we generalise images in the quantale-enriched framework?

Question 2

How do we generalise dilations/erosions in the quantale-enriched framework?

Mathematical Morphology

Definition

Let $\mathbb{1}$ be the $\mathcal{Q}$-category with a single object $\mathbb{1}_0 = \{*\}$ and hom-object $\mathbb{1}(*,*) = 1$. A $\mathcal{Q}$-co-presheaf is a $\mathcal{Q}$-profunctor $\varphi : \mathbb{H} \looparrowright \mathbb{1}$.

Question 2

How do we generalise dilations/erosions in the quantale-enriched framework?

Mathematical Morphology

Definition

Let $\mathbb{1}$ be the $\mathcal{Q}$-category with a single object $\mathbb{1}_0 = \{*\}$ and hom-object $\mathbb{1}(*,*) = 1$. A $\mathcal{Q}$-co-presheaf is a $\mathcal{Q}$-profunctor $\varphi : \mathbb{H} \looparrowright \mathbb{1}$.

Definition

Let $F : \mathbb{D} \to \mathbb{E}$ be a $\mathcal{Q}$-functor and $R : \mathbb{C} \looparrowright \mathbb{D}$ and $S : \mathbb{D} \looparrowright \mathbb{C}$ be $\mathcal{Q}$-profunctors:

- $\underrightarrow{\mathrm{colim}}(R, F) : \mathbb{C} \to \mathbb{E}$ is the unique $\mathcal{Q}$-functor where $\mathbb{E}(\underrightarrow{\mathrm{colim}}(R, F)-, -) = R \blacktriangleright \mathbb{E}(F-, -)$,
- $\underleftarrow{\mathrm{lim}}(S, F) : \mathbb{C} \to \mathbb{E}$ is the unique $\mathcal{Q}$-functor where $\mathbb{E}(-, \underleftarrow{\mathrm{lim}}(S, F)-) = \mathbb{E}(-, F-) \blacktriangleleft S$.

Mathematical Morphology

Definition

Let $\mathbb{1}$ be the $\mathcal{Q}$-category with a single object $\mathbb{1}_0 = \{*\}$ and hom-object $\mathbb{1}(*,*) = 1$. A $\mathcal{Q}$-co-presheaf is a $\mathcal{Q}$-profunctor $\varphi : \mathbb{H} \looparrowright \mathbb{1}$.

Definition

$$\begin{tikzcd}
\mathbb{C} \arrow[dr, "R", "\odot"'] & & \mathbb{E} \\
& \mathbb{D} \arrow[ur, "F"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbb{C} \arrow[rr, dashed, "R \blacktriangleright \mathbb{E}(F-,-)", "\odot"'] \arrow[dr, "R", "\odot"'] & & \mathbb{E} \\
& \mathbb{D} \arrow[ur, "\mathbb{E}(F-,-)"', "\odot"] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbb{C} \arrow[rr, dashed, "\mathrm{colim}(R,F)"] \arrow[dr, "R", "\odot"'] & & \mathbb{E} \\
& \mathbb{D} \arrow[ur, "F"'] &
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{C} & & \mathbb{E} \\
& \mathbb{D} \arrow[ul, "S", "\odot"'] \arrow[ur, "F"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbb{C} \arrow[rr, dashed, "\mathbb{E}(-,F-) \blacktriangleleft S", "\odot"'] & & \mathbb{E} \\
& \mathbb{D} \arrow[ul, "S", "\odot"'] \arrow[ur, "\mathbb{E}(-,F-)"', "\odot"] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbb{C} \arrow[rr, dashed, "\mathrm{lim}(R,F)"'] \arrow[dr, "R", "\odot"'] & & \mathbb{E} \\
& \mathbb{D} \arrow[ur, "F"'] &
\end{tikzcd}$$

Mathematical Morphology

- A $\mathcal{Q}$-category is (co-)complete if it admits all weighted (co-)limits.
- A $\mathcal{Q}$-functor $F : \mathbb{C} \to \mathbb{D}$ is (co-)continuous if it preserves all weighted (co-)limits.

Proposition

Let $\mathbb{C}, \mathbb{D}$ be $\mathcal{Q}$-categories. Then $\langle \mathbb{C}, \mathbb{D} \rangle$ is (co-)complete.

Mathematical Morphology

Lemma [9]

Let $\mathbb{H}$-Dil be the $\mathcal{Q}$-category where:

- $\mathbb{H}\text{-Dil}_0 := \{F : \langle \mathbb{H}, \mathbb{1} \rangle \to \langle \mathbb{H}, \mathbb{1} \rangle \mid F \text{ is co-continuous}\}$,
- $\mathbb{H}\text{-Dil}(F, F') := \bigwedge_{\varphi} \langle \mathbb{H}, \mathbb{1} \rangle (F\varphi, F'\varphi)$.

Then $\mathbb{H}\text{-Dil} \simeq \langle \mathbb{H}, \mathbb{H} \rangle$ in $\mathbf{Cat}(\mathcal{Q})$.

Proof (Outline)

- ($\Leftarrow$) Consider any $R : \mathbb{H} \looparrowright \mathbb{H}$ and let $- \otimes R : \langle \mathbb{H}, \mathbb{1} \rangle \to \langle \mathbb{H}, \mathbb{1} \rangle$ where $(- \otimes R)(\varphi) = \varphi \otimes R$ for all $\varphi \in \langle \mathbb{H}, \mathbb{1} \rangle_0$.

Mathematical Morphology

	Set M.M.	Grey-scale graph M.M.
Colour palette	Boolean Algebra $\mathbb{2}$	Quantale $\mathcal{Q}$
Canvas	Set X	$\mathcal{Q}$-category $\mathbb{H}$
Image	$I \subseteq X$	$\mathcal{Q}$-co-presehaf $\varphi : \mathbb{H} \looparrowright \mathbb{1}$
Relation	$R \subseteq X \times X$	$R : \mathbb{H} \looparrowright \mathbb{H}$
Dilation	$- \oplus R : \mathcal{P}X \to \mathcal{P}X$	$- \otimes R : \langle \mathbb{H}, \mathbb{1} \rangle \to \langle \mathbb{H}, \mathbb{1} \rangle$
Erosion	$- \ominus R : \mathcal{P}X \to \mathcal{P}X$	$- \blacktriangleleft R : \langle \mathbb{H}, \mathbb{1} \rangle \to \langle \mathbb{H}, \mathbb{1} \rangle$

Final Question

Can we express the dilations and erosions induced by the left/right converse in terms of the original $\mathcal{Q}$-profunctor?

Mathematical Morphology

Lemma

Let $R : \mathbb{H} \looparrowright \mathbb{H}$. Then, the following diagrams commute:

$$\begin{tikzcd}
\langle \mathbb{H}, \mathbb{1} \rangle^{op} \arrow[r, "- \blacktriangleleft R"] & \langle \mathbb{H}, \mathbb{1} \rangle^{op} \arrow[d, "\neg"] \\
\langle \mathbb{H}, \mathbb{1} \rangle \arrow[u, "\neg"] \arrow[r, "- \otimes \circlearrowright R"'] & \langle \mathbb{H}, \mathbb{1} \rangle
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \mathbb{H}^{\star}, \mathbb{1} \rangle^{op} \arrow[r, "- \blacktriangleleft R"] & \langle \mathbb{H}^{\star}, \mathbb{1} \rangle^{op} \arrow[d, "\neg^{op}"] \\
\langle \mathbb{H}^{\star}, \mathbb{1} \rangle \arrow[u, "\neg^{op}"] \arrow[r, "- \otimes \circlearrowleft R"'] & \langle \mathbb{H}^{\star}, \mathbb{1} \rangle
\end{tikzcd}$$

$$\begin{tikzcd}
\langle \mathbb{H}, \mathbb{1} \rangle \arrow[r, "- \otimes R"] & \langle \mathbb{H}, \mathbb{1} \rangle \arrow[d, "\neg"] \\
\langle \mathbb{H}, \mathbb{1} \rangle^{op} \arrow[u, "\neg"] \arrow[r, "- \blacktriangleleft \circlearrowright R"'] & \langle \mathbb{H}, \mathbb{1} \rangle^{op}
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \mathbb{H}^{\star}, \mathbb{1} \rangle \arrow[r, "- \otimes R"] & \langle \mathbb{H}^{\star}, \mathbb{1} \rangle \arrow[d, "\neg^{op}"] \\
\langle \mathbb{H}^{\star}, \mathbb{1} \rangle^{op} \arrow[u, "\neg^{op}"] \arrow[r, "- \blacktriangleleft \circlearrowleft R"'] & \langle \mathbb{H}^{\star}, \mathbb{1} \rangle^{op}
\end{tikzcd}$$

(in $\mathbf{Cat}(\mathcal{Q})$) (in $\mathbf{Cat}(\mathcal{Q}^{coop})$)

Thank you for listening!

References I

[1] Ignacio Bellas Acosta and John G. Stell. Monotone Ω-Sup-Fuzzy Relations: Converse and Complementation. In Uli Fahrenberg, Wesley Fussner, and Roland Glück, editors, *Relational and Algebraic Methods in Computer Science*, pages 65–82, Cham, 2024. Springer Nature Switzerland.

[2] Jean Cousty, Laurent Najman, Fabio Dias, and Jean Serra. Morphological filtering on graphs. *Computer Vision and Image Understanding*, 117(4):370–385, April 2013.

[3] Peter Jipsen and Jaš Šemrl. Representable and Diagonally Representable Weakening Relation Algebras. In Roland Glück, Luigi Santocanale, and Michael Winter, editors, *Relational and Algebraic Methods in Computer Science*, pages 140–157, Cham, 2023. Springer International Publishing.

[4] Jean Serra. *Image Analysis and Mathematical Morphology*. Academic Press, Inc., USA, 1983.

References II

[5] Jean Serra, Pierre Soille, and Max A. Viergever, editors. *Mathematical Morphology and Its Applications to Image Processing*, volume 2 of *Computational Imaging and Vision*. Springer Netherlands, Dordrecht, 1994.

[6] John G. Stell. Relations on Hypergraphs. In Wolfram Kahl and Timothy G. Griffin, editors, *Relational and Algebraic Methods in Computer Science*, Lecture Notes in Computer Science, pages 326–341, Berlin, Heidelberg, 2012. Springer.

[7] John G. Stell. Symmetric Heyting relation algebras with applications to hypergraphs. *Journal of Logical and Algebraic Methods in Programming*, 84(3):440–455, May 2015.

[8] John G. Stell, Renate A. Schmidt, and David Rydeheard. A bi-intuitionistic modal logic: Foundations and automation. *Journal of Logical and Algebraic Methods in Programming*, 85(4):500–519, June 2016.

References III

[9] Isar Stubbe. Categorical Structures Enriched in a Quantaloid: Categories, Distributors and Functors, September 2004. arXiv:math/0409473.